Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

1. Tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and use their own software and AI to make them faster and with fewer skilled people than older competitors. They earn money on what they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product. The takeaways below cover why demand is rising, what makes these businesses hard to build, and where the money ends up.

1. **Some tailwinds behind US production are lasting and others can reverse. The lasting ones are what we should build a thesis on.**
   - **Lasting:** China's control of mineral refining and magnets, the shortage of skilled factory workers, and the years it takes a big buyer to approve a new supplier. None of these goes away with an election or a budget cycle.
   - **Temporary:** tariffs, the current surge in munitions spending and interest rates. Tariffs are set by proclamation and can change overnight. Munitions money depends on bills Congress has not passed. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR).
   - **Implications:** we should favor companies that still work if tariffs fall and the munitions surge slows. That means owning approved capacity that buyers need for a decade, and avoiding businesses whose margin exists only because of a tariff.
2. **China's hold on minerals and magnets is the most lasting limit in the theme.**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA). Refining is the step that turns ore into pure metal. China also makes about 94% of the world's rare-earth magnets (Mining.com). These magnets go into drones, missiles, motors and robots.
   - The US is fully import-reliant for 16 critical minerals (USGS). China has already used export controls on these materials, and its current pause on those controls ends on November 10 and November 27, 2026.
   - This is why buyers and the government will pay for a non-Chinese supply even when it costs more. It also means the plants that fix this problem are expensive and slow to build, which takeaway 5 covers.
3. **Defense demand and government money are real, but much of the money is conditional and slow.**
   - The FY27 budget request asks for $49.5B for 10 key munitions, about 8.5 times the past decade's average. About 82% of that depends on a reconciliation bill Congress has not passed (CNAS). The Navy is putting $900M into plants that make submarine parts (Breaking Defense).
   - The FY26 defense bill lets the Pentagon buy munitions over several years and commit to purchases of up to 10 years, with advance payments. That is new and helpful, though each year still needs Congress to approve the money (FY26 NDAA).
   - Government now acts as lender and investor. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to one magnet maker and a $500M conditional loan to a rare-earth metals maker, and about $20B is requested for FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE). Governments worldwide put about $65B into minerals in 2025 (IEA). These loans take a long time to close and come with conditions.
4. **The labor gap is lasting, and it is driving owners of small shops to sell.**
   - US manufacturers need to fill about 3.8M jobs by 2033, and about 1.9M of them may go unfilled (Deloitte).
   - Of 17,156 US machine shops, 83% have fewer than 20 employees (Census). Many owners are near retirement with no one to take over. That creates a supply of certified shops for sale at reasonable prices.

- Fewer skilled people means a plant that needs fewer of them can grow where others cannot. This is the opening for software and automation in takeaway 7.

5. **Capex before contracts is the central risk. A factory has to spend its money before it wins the orders.**
   - Capex means capital spending on buildings, machines and certifications. A big defense contractor, often called a prime, won't approve a new supplier or sign a volume contract until the plant exists and is certified. Approval takes 1-3 years. In expert calls, one prime said 9-12 months and another said one to two years.
   - The numbers are large. One precision-parts maker committed about $200M to a single Arizona plant (Plant Services), and its CEO says capex runs about two years ahead of revenue (claim). Process plants need about 80-85% use of capacity to make money (FTI Consulting). A hardware company at Series B often has about $10M of revenue and break-even gross margins (Lowercarbon).
   - This hurts venture returns. Equity spent on a building earns building-like returns and dilutes founders before the first big order arrives.
   - The winners shift who pays for the build. Options include government loans and grants, a buyer that agrees to purchase output before the plant is built (the Pentagon agreed to buy all of one magnet plant's output for 10 years, with a price floor), a customer that hosts the factory inside its own plant (one 7-year deal), bank debt once revenue exists, leased buildings, franchising, and buying plants that are already certified. One roll-up founder said they buy shops at about seven times annual earnings (claim).
6. **Buyers pay for speed and spare capacity. Few pay for a lower price, so the margin goes to whoever owns the output and holds the approvals.**
   - "Cost is secondary. Cost is driven usually by schedule," a director at a large defense contractor said in an expert call. An engineer there would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - Owning the plant pays. One listed parts maker earns a 49% gross margin in its own plants and 31% on work it farms out to partner shops (company filings). Approved incumbents earn well too: a large engine-parts unit runs a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization). Aerospace programs can run about 50 years, so an approval keeps paying.
   - Selling software or machines to factories and mines has been hard. One 3D-printer maker fell from a $2.1B value to a $42.5M sale (3D Printing Journal). An AI-run machining startup in Germany went insolvent in August 2026. Buyers cap each supplier at about a quarter to a third of their work, so even a winner grows one share at a time.
7. **AI's edge today is in the work around the machine and in processes that run nonstop. Nobody has shown yet that it makes each unit cheaper as volume grows.**
   - Around the machine means quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. One startup claims it cut assembly-instruction writing from about 10 hours to 15 minutes (claim). A former employee of one parts maker said its time from order to first part fell about 50% over two years.
   - The machines themselves are underused. In a survey of more than 100 plants, cutting machines ran about 34% of the time against a 70% target. European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). Better planning and control software closes that gap.
   - In a plant that runs nonstop, like a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. A head of strategy at a large aircraft maker put net savings from automation, after paying for the machines, at 20-30% at best. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case rests on speed and yield more than headcount.
8. **The key risk is that incumbents automate and buy faster than new companies can grow.**

- The primes decide who is approved and how much work each supplier gets. The largest aerospace parts supplier saw its share of one buyer's machining fall from about 50% to 30%, and it is now automating (expert call).
- Big companies buy instead of build. Miners back startups through their venture arms, one paid $6.7B for a lithium producer, and a defense contractor's venture arm led a round in an AI materials startup. Drug companies are building their own AI labs, one with plans for up to $1B.
- There is still room. The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that base is small, old and short of people, which is where new companies can win.

2. Interesting categories

The categories below are grouped by where they sit in the supply chain, from parts to plants to materials. For each, we explain why it is interesting and how the business makes money. We rank parts and fabrication highest because demand is strongest there and the capital needed is smallest. Magnets are last because they need the most capital before contracts.

A. Defense and aerospace parts

How it ties to takeaways 3, 5 and 6: takeaway 3 says defense demand is rising, takeaway 5 says the plant has to exist before the contract, and takeaway 6 says buyers pay for speed and keep paying once a supplier is approved. Parts makers sit where all three meet.

Factories here come in two models. An owned factory is run by the company on its own machines, so it keeps the whole margin and all the data but pays the full cost of building before the orders come. A hybrid model mixes owned plants with capacity someone else paid for, through franchisees, a customer hosting the cells or plants bought from retiring owners. Hybrid companies keep less of each sale but need far less capital. We prefer franchised and acquired-plant models over building every plant from scratch.

Precision parts for defense and aerospace

- **Why it's interesting:** Precision parts are metal parts made to very tight limits, by computer-controlled machining (CNC), metal 3D printing or casting. Missiles, jets, satellites and submarines need thousands of them. The shops that make them are small and slow, and a late part holds up a whole program. A new supplier that delivers in weeks gets paid a premium and, once approved, keeps the work for years.
- **How the business makes money:** The company sells finished parts, usually to a prime or a large supplier under a purchase order or a multi-year contract. The buyer sends a drawing and gets parts back. Margin comes from charging for speed and running machines more hours a day with fewer people, using software for quoting, programming and inspection. Revenue grows as more part numbers get approved.
- **What it costs to build and the factory model:** An owned plant costs tens to hundreds of millions of dollars before approval; one parts maker committed about $200M to a single plant, and the Navy co-funds some submarine-part plants. Some owned operators also place cells inside a customer's plant, which shares the cost. A franchise network flips the cost: one network claims 14 factories, 4 owned and 10 franchised, where franchisees buy the machines and the company supplies software, orders and quality rules (claim). It also claims it cut one customer's lead time from about 10 weeks to under 14 days (claim). The question for franchising is whether primes accept the same quality across many sites.

Buying and automating certified shops

- **Why it's interesting:** Takeaway 4 says thousands of small shop owners are retiring. Their plants already hold the hard part: certifications and approvals on parts that are often single-source, meaning only one supplier is approved. Buying a plant skips the 1-3 year approval wait in takeaway 5.

- **How the business makes money:** The company buys a profitable, certified shop, keeps its customers and contracts, and adds software and automation to raise output and margin. It earns the shop's existing profit from day one and grows it. Certified here usually means AS9100, the quality standard for aerospace and defense suppliers.
- **What it costs to build and the factory model:** This is owned by acquisition. The machines are already paid for, so build risk is low. The risk moves to the purchase price, the debt used to pay it and whether key engineers stay. One roll-up founder said they pay about seven times annual earnings (claim). One platform owns eight acquired certified plants making parts for fighter jets, torpedoes and tanks.

B. Fabrication and assembly

How it ties to takeaways 1 and 6: takeaway 1 says we want demand that lasts past tariffs, and takeaway 6 says buyers pay for speed. Fabrication serves defense and every other kind of hardware company, and the best operators turn reorders in about a day.

Fabrication means cutting, bending and welding metal into frames, brackets and enclosures, plus assembling parts like wire harnesses. The work is high-mix, meaning many small orders that are each a bit different. Plants cost far less than precision or materials plants, so the capex risk is smaller.

Custom parts and sheet metal

- **Why it's interesting:** Hardware startups, robot makers, energy companies and defense suppliers all need custom metal parts fast. Most shops quote by hand and take days to answer. Fabricated-metal shipments grew from $471.5B in 2023 to $490.4B in 2025 (Census), though only 14 of 64 tariff-linked reshoring announcements had become real projects by mid-2026 (Reshoring Initiative).
- **How the business makes money:** Customers upload a design online, get an instant price and receive parts in days. The company earns a margin on each order and wins on speed and convenience. One online fabricator handles about 32,000 orders a month and says it has been profitable for eight years (claim). Its founder says he beats China on price on only about 20-25% of his work (claim), so speed carries the rest.
- **What it costs to build and the factory model:** Usually owned plants with lasers, press brakes and finishing lines, costing a fraction of a precision plant. A newer approach uses pairs of robot arms to shape sheet metal without the costly dies a stamping press needs, and sometimes places robot cells at customer sites, which makes it hybrid. Each part takes minutes to hours against seconds on a press (MMS), so it wins on short runs and design changes. The risk is that customers outgrow online fabricators once their volumes rise.

Wire harnesses and assembly

- **Why it's interesting:** A wire harness is the bundle of wires and connectors that carries power and signals through an aircraft, satellite or missile. Most are still built by hand, and a late harness holds up the whole system. One maker claims it cut harness lead time from six months to two weeks (claim).
- **How the business makes money:** The company sells finished harnesses and cable assemblies per program. Software turns the customer's design into a digital model and step-by-step instructions for technicians, which raises the share that pass inspection the first time and lets less experienced workers do the job. Margin comes from fewer reworks and faster delivery.
- **What it costs to build and the factory model:** Owned plants, but light ones, since the work is mostly benches, testers and people. One maker leased a second plant of about 80,000 square feet and opened it within months. Leasing is the kind of light capex answer to takeaway 5 we want.

C. Software that runs a plant, paid on results

How it ties to takeaways 5 and 7: takeaway 5 says plants are expensive to build, and takeaway 7 says AI's clearest edge is in controlling processes that run nonstop. Software that runs an existing plant earns from the plant without paying to build one. It is the closest fit for an enterprise software fund.

Takeaway 6 warns that selling software to traditional plants has been hard. This category differs on two points: the software adjusts the plant directly, and it is paid on the extra output it creates.

Plant control software

- **Why it's interesting:** Chemical, metals and other process plants run below the capacity they need, about 74% in Europe against 80-85% (FTI Consulting). In a sold-out plant, each extra point of throughput or yield is close to pure profit. The old control-system vendors mostly sell hardware or advise operators, which leaves room for software that acts.
- **How the business makes money:** The software connects to the plant's existing controls, adds cheap sensors such as cameras and microphones, builds a live model of the plant and adjusts setpoints itself through a safety layer. A setpoint is a target such as a temperature or a flow rate. One startup says it is paid on the production it adds (claim). That makes revenue a share of new profit, so the buyer sees the value before it pays.
- **What it costs to build and the factory model:** No factory. The cost is engineering, sensors and long deployments at each plant. The hard parts are agreeing on a fair baseline for results-based pricing and getting a plant's safety team to approve software that makes changes on its own. Buyers are large industrial companies, where our go-to-market and partnership help counts.

D. Automated chemistry labs for pharma

How it ties to takeaways 1, 6 and 7: takeaway 1 lists drug-supply security as a lasting push, takeaway 6 says owning the output keeps the margin, and takeaway 7 says AI does best where every run is measured. Robotic chemistry labs meet all three. They are early and still need paying customers.

Drug companies send most early chemistry work offshore, largely to China. The two biggest offshore providers are worth close to $100B combined (Lightspeed). BIOSECURE, a law passed in December 2025, limits US drug companies' use of certain Chinese biotech suppliers. A survey found 79% of 124 biotechs had a China-based contract manufacturer, and replacing one can take up to eight years (BIO).

Robotic labs that make new molecules

- **Why it's interesting:** These labs turn molecule designs into code that robots run, then log every result, including failures. Failed reactions are data most labs throw away, and they make the next prediction better. A US or UK option is attractive to buyers worried about Chinese suppliers.
- **How the business makes money:** Today they sell discovery chemistry as a service, making small batches of new compounds for drug programs, paid per project or per molecule. Over time some plan to sell predictions about which molecules will work. One lab claims it works with eight of the top 20 drug companies (claim). Most output is milligrams to grams, so this is research work for now.
- **What it costs to build and the factory model:** Owned labs. Each lab costs tens of millions of dollars, and some have pushed part of that onto governments; one received about £22M of UK and Scottish grants toward its next lab. The risk is that big drug companies build their own robotic labs instead of buying from startups. Until labs show named paying customers at volume, most are not investable now.

E. Selective: rare-earth metals and magnets

How it ties to takeaways 2 and 5: takeaway 2 says China's hold on magnets is the most lasting limit, and takeaway 5 says these plants carry the most capex before contracts. We would back at most one company here.

We would invest only where government debt pays for most of the plant and a buyer has signed to purchase the output. The only scaled US magnet producer still lost $85.9M on $224.4M of revenue in 2025 (company filings).

Rare-earth metals and magnets

- **Why it's interesting:** Rare-earth magnets, mostly the neodymium-iron-boron type, power drones, chip tools, robots and electric motors. Buyers need a non-Chinese source, and the Pentagon has shown it will sign long purchase deals with a price floor to get one.
- **How the business makes money:** Metal makers turn refined rare-earth oxides into metal; magnet makers turn the metal into magnets. Both sell under long contracts to defense, auto and electronics buyers. Margins depend on beating Chinese prices or on a guaranteed price floor.
- **What it costs to build and the factory model:** Owned plants, and very expensive ones. One magnet maker is building a 10,000-ton plant with a $620M government loan and no commercial buyer named. A rare-earth metals maker announced a $375M plant with one $100M automaker contract and plans to grow about 15x in two years. These are the clearest cases of capex before contracts, which is why we stay selective.